

AARUSH RASTOGI

B. Tech, SRMIST 2027

Email ID: aarushras18@gmail.com

Contact: +91-9625726211

[LinkedIn](#) | Website: [Portfolio](#)

PROFESSIONAL SUMMARY

Computer Science Engineering undergraduate (B.Tech, 2027) with a strong foundation in Data Structures & Algorithms, machine learning, and software development. Passionate about teaching and mentoring peers, with experience explaining technical concepts and problem-solving approaches. Proven leadership through a campus technical club, with experience organising DSA workshops, coding bootcamps, and mentoring sessions. Combines deep problem-solving skills with the ability to communicate reasoning clearly — driven by the belief that the best engineers are also exceptional teachers.

SKILLS

Programming Languages: Python, SQL, C++ Programming, Java (Basic)

Software Development: HTML, CSS, DBMS, Data Structures & Algorithms, Competitive Programming

Machine Learning & Analytics: Machine Learning, Prediction Modelling, Data Analysis, Data Visualization

Tools & Technologies: TensorFlow, TensorFlow Lite, Tkinter, Matplotlib

Professional Skills: Problem Solving, Algorithmic Thinking, Mentoring & Peer Teaching, Content Development, Leadership, Communication, Team Collaboration.

RELEVANT COURSEWORK

- Data Structures & Algorithms
- Database Management Systems (DBMS)
- Object-Oriented Programming (OOP)
- Operating Systems
- Machine Learning
- Data Analytics
- Probability & Statistics
- Software Engineering

PROJECTS

Self-Care Synergy

Jan 2025

- Developed a self-care management platform that provides daily scheduling, health insights, and caregiver assistance.
- Designed and implemented user authentication and data management functionality using SQL.
- Built a user-friendly web interface using HTML and CSS to improve accessibility and user experience.
- Developed features to organize user information and support personalized recommendations based on user inputs.
- Focused on creating a structured and scalable solution for managing self-care activities and related data.

House Price Prediction Model

May 2025

- Developed a GUI-based application using Python, Tkinter, and Linear Regression to estimate housing prices based on property attributes.
- Processed and analyzed housing data to understand the relationship between key features and property values.
- Implemented real-time prediction functionality with automated CSV-based data storage and management.
- Created visualizations using Matplotlib to support analysis and interpretation of housing price trends.
- Tested and evaluated model predictions to ensure consistency and reliability of results.

VisionDX – Retinal Disease Detection System

Aug 2025

- Developed a deep learning-based system using CNNs and TensorFlow to detect and classify multiple retinal diseases from fundus images.
- Built an end-to-end workflow including image preprocessing, model training, and prediction generation for disease classification.
- Worked with image datasets to improve model performance and ensure accurate disease detection.
- Evaluated model performance using classification metrics to assess prediction effectiveness and reliability.
- Designed the system to support automated diagnosis by providing faster and more consistent disease identification.

EdgeVisionNet: Live Energy Intelligence Edge AI Platform

Jan 2026

- Developed a full-stack Edge AI platform to monitor device telemetry, including CPU usage, memory consumption, battery status, and energy utilization during AI inference.
- Built a lightweight image classification model using MobileNetV2 with an attention mechanism for efficient on-device image processing.
- Implemented TensorFlow Lite optimization and API-based inference to improve deployment efficiency and reduce latency.

- Designed and developed an interactive web dashboard to visualize system performance, model behavior, and energy consumption metrics.
- Evaluated multiple lightweight models by comparing accuracy, latency, and energy efficiency to understand performance trade-offs.
- Integrated real-time monitoring and analytics features to support performance tracking and informed decision-making.

PROFESSIONAL CERTIFICATIONS

Machine Learning A-Z	Jul 2025
Python Pro	May 2025
AI & ML Bootcamp	May 2025

EDUCATION

B. Tech, SRMIST	2023 - 2027
XII (Science), Airforce School Ghaziabad	2022 - 2023
X, Airforce School Ghaziabad	2020 - 2021

AWARDS & ACTIVITIES

- Awarded **NCC "A" Certificate**, showcasing commitment to physical fitness, discipline, and community service in March 2021.
 - Head Boy – Led the student council, coordinated events, mediated between students and faculty, and represented the school in interschool functions for session 2022-23.
 - Managed several college events and led a team of 150-200 volunteers towards the success of the event.
 - **Co-Executive, CSI-SRM NCR Chapter(2024-Present)** – Organized technical workshops, coding bootcamps, hackathons, and coding competitions reaching 2000+ students. Guided junior members during technical events and peer-learning sessions, helping them strengthen problem-solving and programming fundamentals.
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